

Appendix

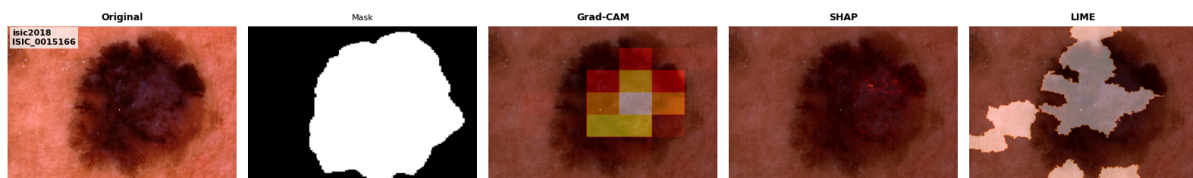
DermaXplain: Evaluating Explainable AI Methods Against Clinically Structured Pseudo-Concepts in Skin Lesion Classification

Thierry Beliere Janavi Ravi

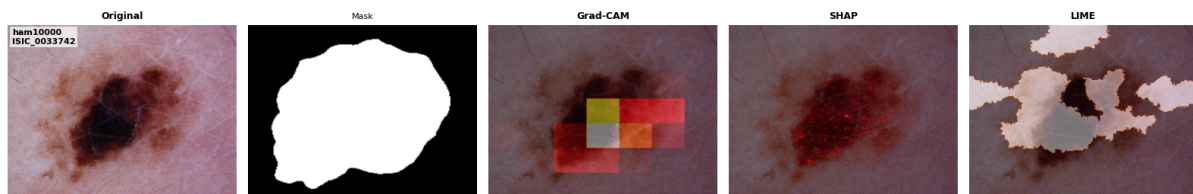
Section 1 - Qualitative Examples

1.1 - Examples of XAI Saliency Maps

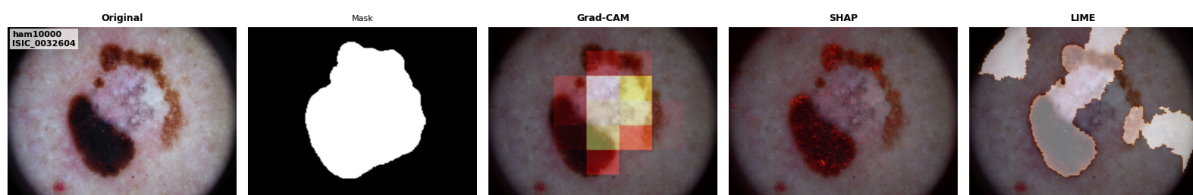
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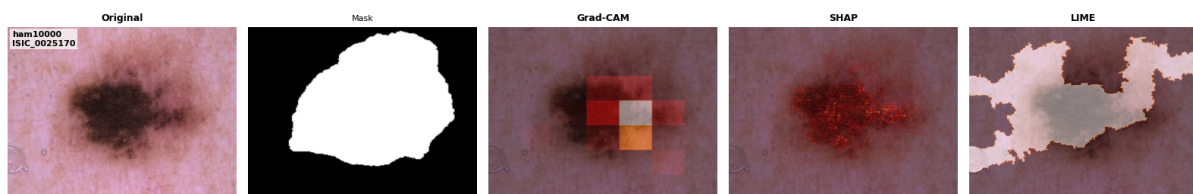
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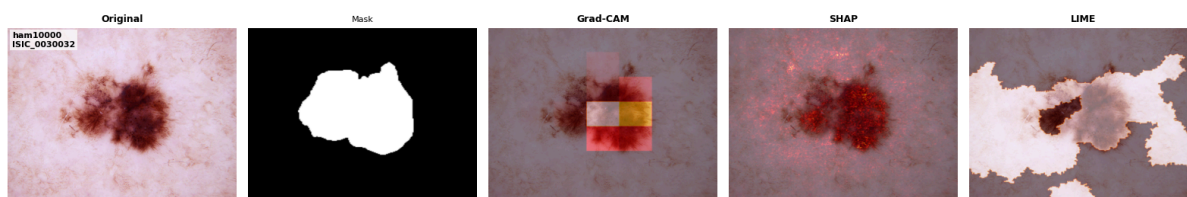
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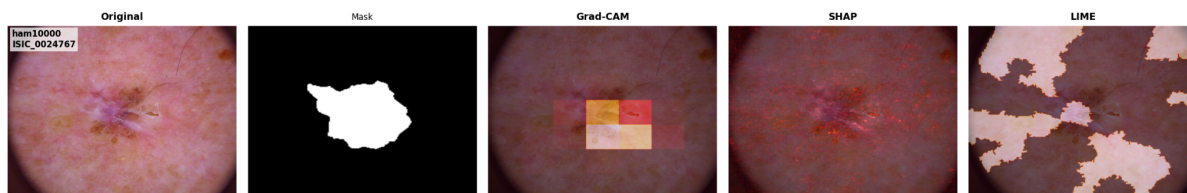
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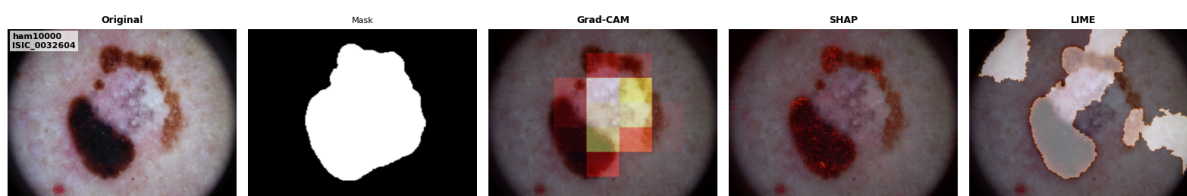
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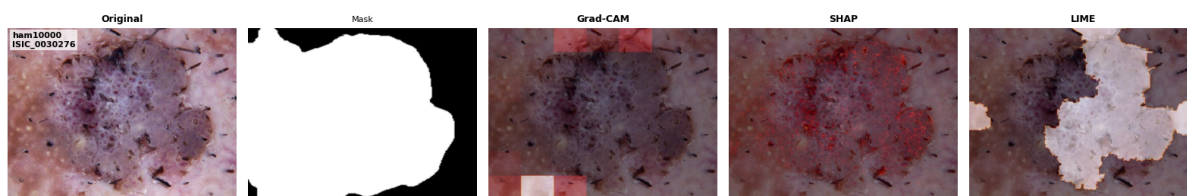
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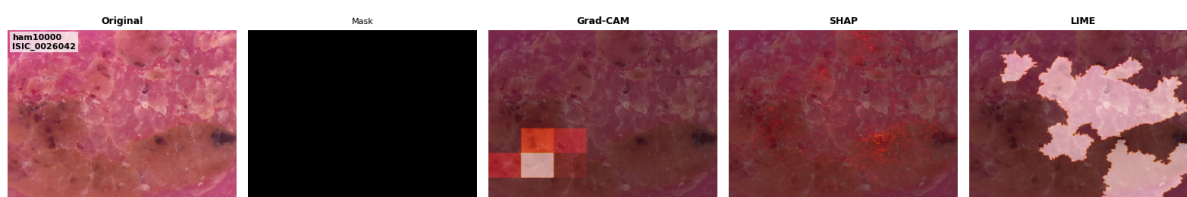
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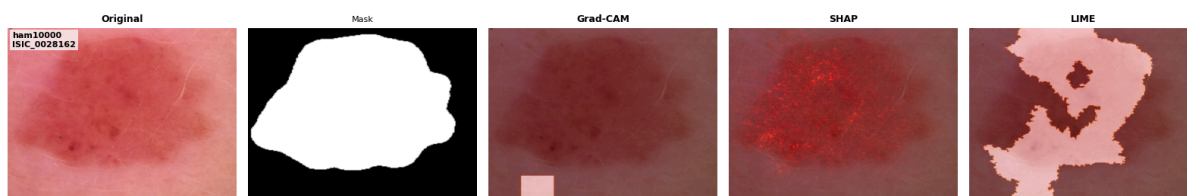
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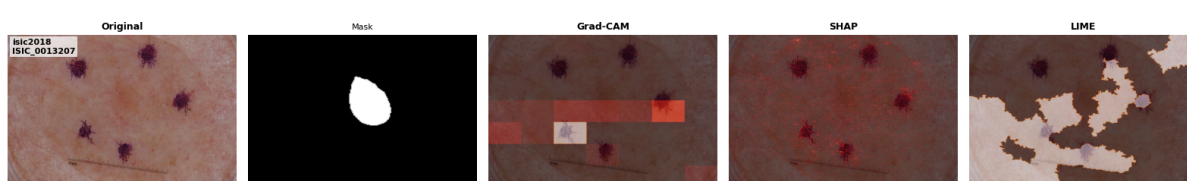
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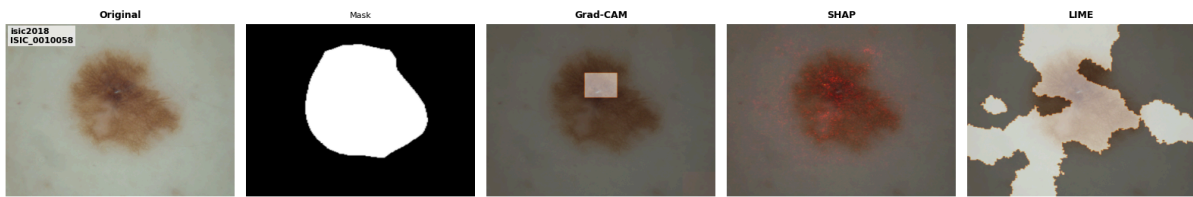
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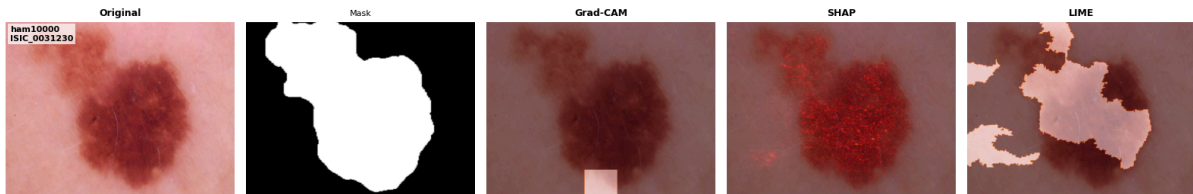
isic2018 / ISIC_0013207



isic2018 / ISIC_0010058

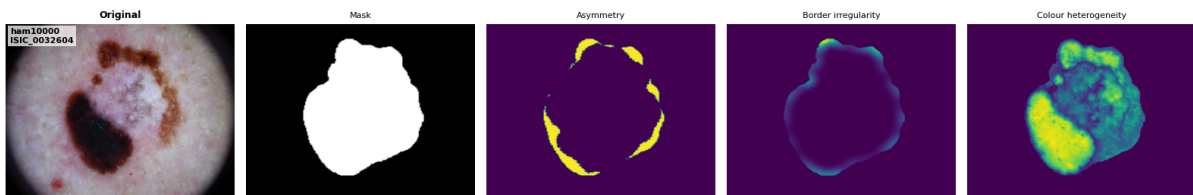


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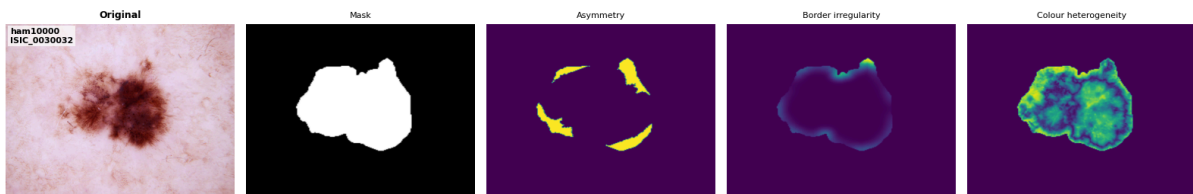


1.2 - Examples of Pseudo-concept Maps

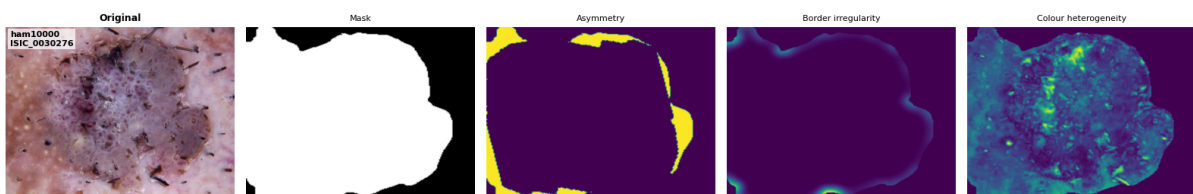
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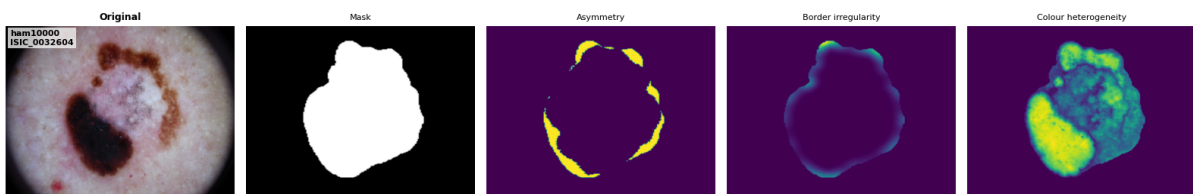
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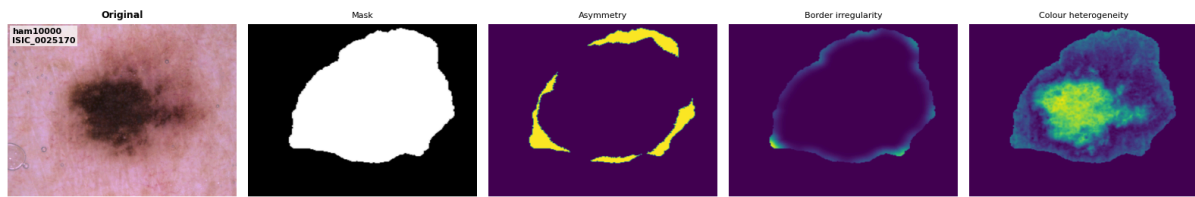
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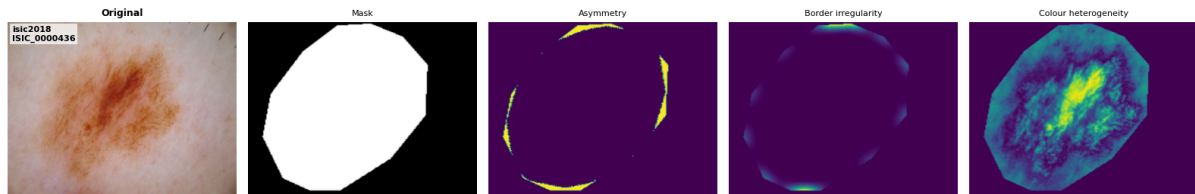
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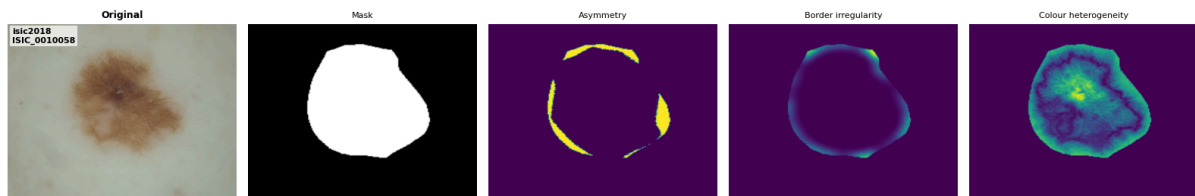
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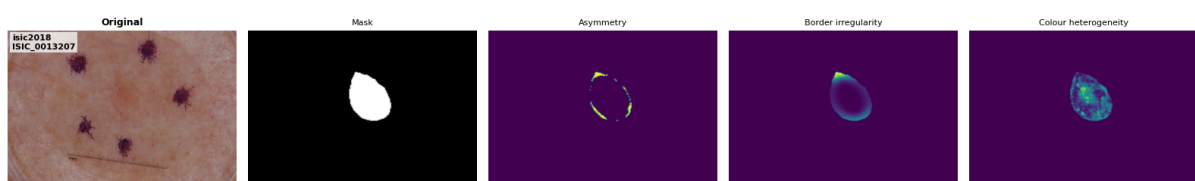
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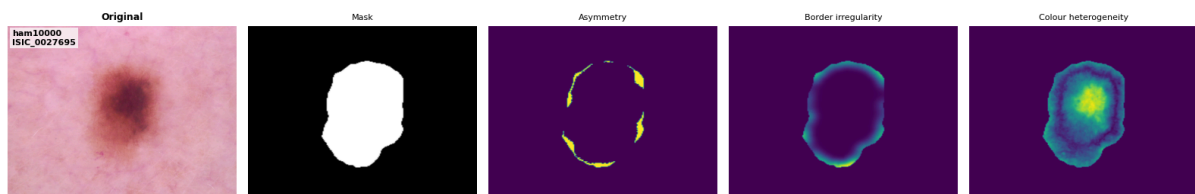
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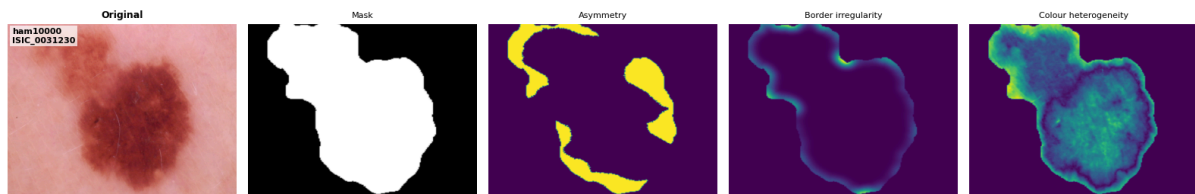
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ham10000 / ISIC_0031230



Section 2 - Metrics formulas

All summations are taken over the spatial pixel indices (i, j) of the saliency map and reference mask.

Name / Definition	Formula	Interpretation
H: Saliency Map: Normalised continuous saliency map generated by an XAI method.	$H = \{H_{i,j} \mid 0 \leq H_{i,j} \leq 1\}$	k was set at 20% during this research.
M: Lesion mask: Binary reference mask representing either the lesion or a pseudo-concept region.	$M_{i,j} \in \{0, 1\}$	
X: Binary top-k% saliency region derived from the continuous saliency map H	$X_{i,j} \in \{0, 1\}$ where: $X_{i,j} = \mathbf{1}[H_{i,j} \geq \tau_k(H)]$ and: $\tau_k(H) = 1 - k$	
Intersection over Union; measures overlap between the binary top-k% XAI region and the reference mask relative to their union.	$IoU(X, M) = \frac{\sum_{i,j} X_{i,j} M_{i,j}}{\sum_{i,j} (X_{i,j} + M_{i,j} - X_{i,j} M_{i,j})}$	0 = no overlap 1 = identical binary regions
Dice: Measures similarity between the binary top-k% XAI region and the reference mask, giving greater weight to overlap.	$Dice(X, M) = \frac{2 \sum_{i,j} X_{i,j} M_{i,j}}{\sum_{i,j} X_{i,j} + \sum_{i,j} M_{i,j}}$	0 = no overlap 1 = identical binary regions
SIR (Saliency Inside Ratio); proportion of total continuous saliency mass located inside the reference region.	$SIR(H, M) = \frac{\sum_{i,j} H_{i,j} M_{i,j}}{\sum_{i,j} H_{i,j}}$	0 = no saliency inside 1 = all saliency mass inside
$TIxAI_{norm}(H, M)$: Normalised relevance score comparing mean continuous saliency inside and outside the reference region.	$TIxAI_{norm}(H, M) = \frac{RD_{In}}{RD_{In} + RD_{Out} + \epsilon}$	<0.5: outside dominant = 0.5: equal density > 0.5: inside dominant
RD_{In} : Mean continuous saliency within the reference region	$RD_{In} = \frac{\sum_{i,j} H_{i,j} M_{i,j}}{\sum_{i,j} M_{i,j}}$	
RD_{Out} : Mean continuous saliency outside the reference region	$RD_{Out} = \frac{\sum_{i,j} H_{i,j} (1 - M_{i,j})}{\sum_{i,j} (1 - M_{i,j})}$	

Section 3 - Additional results

3.1 Complete size-stratified metric results

Concept	Size Tier	Method	Mean IoU	Mean Dice	Mean SIR	Mean TlxAl
Asymmetry	Normal	Grad-CAM	0.0311	0.0590	0.0241	0.2411
Asymmetry	Normal	LIME	0.0333	0.0632	0.0369	0.5212
Asymmetry	Normal	SHAP	0.0341	0.0650	0.0376	0.5470
Asymmetry	Small	Grad-CAM	0.0053	0.0104	0.0094	0.2928
Asymmetry	Small	LIME	0.0066	0.0131	0.0067	0.6057
Asymmetry	Small	SHAP	0.0120	0.0235	0.0088	0.6912
Asymmetry	Tiny	Grad-CAM	0.0025	0.0049	0.0058	0.2965
Asymmetry	Tiny	LIME	0.0032	0.0064	0.0033	0.6353
Asymmetry	Tiny	SHAP	0.0057	0.0113	0.0044	0.7223
Asymmetry	Very Tiny	Grad-CAM	0.0016	0.0031	0.0032	0.2478
Asymmetry	Very Tiny	LIME	0.0017	0.0034	0.0018	0.6022
Asymmetry	Very Tiny	SHAP	0.0033	0.0066	0.0026	0.7344
Border Irreg.	Normal	Grad-CAM	0.0060	0.0119	0.0045	0.1810
Border Irreg.	Normal	LIME	0.0067	0.0132	0.0068	0.4552
Border Irreg.	Normal	SHAP	0.0080	0.0158	0.0070	0.5144
Border Irreg.	Small	Grad-CAM	0.0073	0.0144	0.0126	0.2627
Border Irreg.	Small	LIME	0.0102	0.0201	0.0104	0.6011
Border Irreg.	Small	SHAP	0.0175	0.0344	0.0130	0.6887
Border Irreg.	Tiny	Grad-CAM	0.0066	0.0129	0.0155	0.2946
Border Irreg.	Tiny	LIME	0.0095	0.0188	0.0098	0.6425
Border Irreg.	Tiny	SHAP	0.0157	0.0309	0.0122	0.7188
Border Irreg.	Very Tiny	Grad-CAM	0.0014	0.0028	0.0058	0.0714
Border Irreg.	Very Tiny	LIME	0.0014	0.0027	0.0014	0.1081
Border Irreg.	Very Tiny	SHAP	0.0028	0.0055	0.0023	0.1346

The four metrics respond differently to lesion size and pseudo-concept sparsity. Binary overlap generally decreases for smaller lesions, while TlxAl may increase when saliency becomes more concentrated within a small target region.

3.2: Colour heterogeneity by lesion-size tier

Size Tier	Method	Mean IoU	Mean Dice	Mean SIR	Mean TlxAI
Normal	Grad-CAM	0.0545	0.0990	0.0629	0.3217
Normal	LIME	0.0587	0.1071	0.0652	0.5798
Normal	SHAP	0.0854	0.1514	0.0828	0.6609
Small	Grad-CAM	0.0121	0.0234	0.0295	0.2921
Small	LIME	0.0173	0.0337	0.0178	0.6301
Small	SHAP	0.0281	0.0541	0.0238	0.7290
Tiny	Grad-CAM	0.0064	0.0126	0.0160	0.2795
Tiny	LIME	0.0096	0.0189	0.0098	0.6326
Tiny	SHAP	0.0160	0.0314	0.0143	0.7520
Very Tiny	Grad-CAM	0.0036	0.0070	0.0089	0.2441
Very Tiny	LIME	0.0042	0.0083	0.0043	0.5710
Very Tiny	SHAP	0.0087	0.0172	0.0079	0.7584

Colour-heterogeneity overlap decreases with lesion size, but SHAP TlxAI increases from normal to very tiny lesions. This divergence shows that absolute overlap and relative saliency concentration capture different aspects of alignment.

3.3 Chance-adjusted Dice enrichment by size tier

Concept	Size Tier	Method	Raw Dice	Dice Enrichment
Asymmetry	Normal	Grad-CAM	0.059	0.963
Asymmetry	Normal	LIME	0.063	0.793
Asymmetry	Normal	SHAP	0.065	0.596
Asymmetry	Small	Grad-CAM	0.010	1.093
Asymmetry	Small	LIME	0.013	0.960
Asymmetry	Small	SHAP	0.024	0.947
Asymmetry	Tiny	Grad-CAM	0.005	1.128
Asymmetry	Tiny	LIME	0.006	1.031
Asymmetry	Tiny	SHAP	0.011	0.980
Asymmetry	Very Tiny	Grad-CAM	0.003	0.998
Asymmetry	Very Tiny	LIME	0.003	1.316
Asymmetry	Very Tiny	SHAP	0.007	1.000
Border Irreg.	Normal	Grad-CAM	0.012	0.981
Border Irreg.	Normal	LIME	0.013	0.769
Border Irreg.	Normal	SHAP	0.016	0.509
Border Irreg.	Small	Grad-CAM	0.014	1.009
Border Irreg.	Small	LIME	0.020	0.909
Border Irreg.	Small	SHAP	0.034	0.929
Border Irreg.	Tiny	Grad-CAM	0.013	1.065
Border Irreg.	Tiny	LIME	0.019	0.971
Border Irreg.	Tiny	SHAP	0.031	0.960
Border Irreg.	Very Tiny	Grad-CAM	0.015	0.998
Border Irreg.	Very Tiny	LIME	0.015	0.888
Border Irreg.	Very Tiny	SHAP	0.030	1.013
Colour Heterog.	Normal	Grad-CAM	0.099	1.056
Colour Heterog.	Normal	LIME	0.107	0.986
Colour Heterog.	Normal	SHAP	0.151	1.066
Colour Heterog.	Small	Grad-CAM	0.023	1.014
Colour Heterog.	Small	LIME	0.034	0.996
Colour Heterog.	Small	SHAP	0.054	1.043
Colour Heterog.	Tiny	Grad-CAM	0.013	0.998
Colour Heterog.	Tiny	LIME	0.019	0.956
Colour Heterog.	Tiny	SHAP	0.031	1.034
Colour Heterog.	Very Tiny	Grad-CAM	0.007	0.998
Colour Heterog.	Very Tiny	LIME	0.008	1.069
Colour Heterog.	Very Tiny	SHAP	0.017	1.032

Raw Dice generally decreases for smaller lesions, whereas enrichment often remains near the random baseline. Very tiny lesions should be interpreted cautiously because extremely small observed and random overlaps can make enrichment ratios unstable.

3.4 SHAP Colour-Heterogeneity Enrichment by Diagnostic Group

Diagnostic group	Dice enrichment	SIR enrichment
Melanoma	1.18	1.25
Non-Melanoma	1.04	1.10

Both melanoma and non-melanoma lesions exceeded the area-matched random baseline, with higher Dice and SIR enrichment for melanoma lesions. This indicates stronger-than-random spatial alignment with the colour-heterogeneity proxy, but does not establish causal reliance by the classifier.

Taken together, these supplementary results support the use of multiple complementary metrics rather than a single alignment measure.